



Durability of fiber reinforced polymer (FRP) in simulated seawater sea sand concrete (SWSSC) environment

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ABSTRACT

This paper presents an experimental investigation on the degradation of carbon/glass/basalt fiber reinforced polymer (i.e., CFRP/BFRP/GFRP) exposed to simulated seawater sea sand concrete environments (SWSSC) at 25, 40 and 60 °C for 6 months. The presence of NaCl in simulated concrete environment was found beneficial for the moisture uptake of CFRP and GFRP. The greater fiber degradation of BFRP was attributed to its high aluminium, iron and magnesium contents on fibers. Further, FRPs showed greater degradation resistance in high performance concrete solutions that have a lower alkaline content. Thus, CFRP exhibited the best durability to simulated SWSSC environments, followed by GFRP and BFRP.

1. Introduction

According to a report by United Nations, global population is expected to increase by 2.4 billion by 2050 [1], placing huge demand on the infrastructure, and hence, on their construction materials. As a result, there will be increasing shortage of fresh water and river sand, the two main ingredients of conventional concrete. In addition, there has been an increasing need of additional coastal infrastructures due to the rising sea level [2], requiring large quantities of river sand and fresh water transported from inland to the site. The potential use of seawater and sea sand concrete (SWSSC) is a hugely attractive proposition as substitute for conventional concrete that requires fresh water and river sand.

Though SWSSC offers enormous benefits, it may be nearly impossible to use the traditional carbon steels to act as external confinement or internal reinforcement that will suffer severe corrosion due to very high Cl^- content of seawater and sea sand. ‘Concrete Cancer’, which is spallation and cracking of concrete due to enhanced corrosion of steel reinforcement by Cl^- ions, and hence a serious concern even for conventional concrete, will only be highly accelerated in the presence of seawater and sea sand. A series of studies investigating the corrosion of steels in SWSSC concrete has been summarised in [3]. To mitigate the critical problem of unacceptably high corrosion rate of steels in SWSSC, fiber reinforced polymer (FRP) is proposed as a potential alternative material for SWSSC reinforcements, given their superior corrosion resistance and competent mechanical properties.

Given the distinct potential advantages of FRPs over steels as external

confinement or internal reinforcement material, it is mandatory to ascertain their durable performance in the relatively new environment of SWSSC. A number of studies have explored pre-exposure of FRPs to simulated conventional concrete environments, followed by mechanical testing to characterise deterioration in mechanical properties as a result of the pre-exposure [4–10]. It has been found that concrete environment, the most common source of alkali, is detrimental to the mechanical properties of FRPs while the salt content only has a marginal effect [4,6,8,10,11]. For instance, Chen et al. [6] reported that GFRPs that were exposed to simulated seawater at 60 °C for 70 days exhibited highest tensile strength retention (i.e., 74%), followed by those exposed to distilled water (71%) and then highly alkaline solution (64%), while CFRP retained 96% of its tensile strength after exposure to the most aggressive environment, i.e., the highly alkaline solution at 60 °C. Similar results were reported by [11] for GFRP, BFRP and CFRP after 60 days of immersion in seawater, alkaline solution and distilled water. To explain the difference in the extent of the reported mechanical property losses in the three solution types, some studies have looked into the degradation specifically at fiber, resin and interface during exposures to these solutions. It was reported in [12] that vinyl ester resin of GFRP plasticized during exposure to distilled water, simulated concrete solution and concrete leachate solution. Hydrolytic degradation and fiber degradation were enhanced particularly in the alkaline solution. Similar findings were also reported for epoxy-based BFRP [11]. However, there are very few studies on the durability of FRPs exposed to SWSSC environment, which has a mix of alkali and chloride. El-Hassan et al. [13] found matrix hydrolysis, increasing void content,

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weakening of fiber/matrix interface but no detectable fiber degradation on GFRP bars that were embedded in seawater-contaminated concrete for 15 months. On the other hand, Wang et al. [14] observed chemical attack of fiber in the cross sections of both GFRP and BFRP bars pre-exposed to simulated SWSSC solutions and a greater degradation of fiber/matrix interface in the case of BFRP than GFRP upon exposure to the highly alkaline solution at 55 °C for 63 days. However, it is argued that the degradation at the fiber/matrix interface seen in this study [14] might have been caused (or accentuated) by grinding of cross-section during specimen preparation. Therefore, there might be a need for examining such degradation by an alternative approach.

This study aims to systematically investigate the degradation mechanism of three different types of filament-wound FRPs (i.e. CFRP, GFRP, BFRP) exposed to simulated seawater and sea sand concrete (SWSSC) environments and simulated conventional concrete environments. The results will provide a critical understanding of FRP degradation in SWSSC that will enable a suitable reinforcing material selection.

2. Experimental procedure

2.1. Materials and specimens

Filament-wound CFRP, BFRP and GFRP tubes with a nominal outer diameter of 50 mm and wall thickness of 3.5 mm were procured from CST Composites Australia. The winding pattern is 20% 15° + 40% 45° + 40% 75°, and the epoxy resin in each FRP was a bisphenol A and bisphenol B blend with an amine hardener. Inductively coupled plasma-atomic emission spectroscopy (ICP-AES) was used to determine the chemical composition of intact E-glass and basalt fibers. Table 1 provides the chemical compositions of E-glass and basalt fibers that were present in FRPs are used in the present work.

FRP tubes were cut into coupons (seen in Fig. 1) with dimensions of 30 × 40 × 3.5 mm using a diamond saw. The weight of coupons complied with the weight requirement specified in ASTM D5229 [15]. The cut edges of coupons were sealed with epoxy, and they were cleaned with distilled water, oven dried at 60 °C for 24 h and labelled for the immersion test.

2.2. Test solutions and immersion procedure

In the present work, four different solutions were used to simulate the pore solutions of normal concrete (NC), high performance concrete (HPC), seawater and sea sand normal concrete (SWSSNC), seawater and sea sand high performance concrete (SWSSHPC). Distilled water (DW) was used as a reference. Chemical compositions and pH of these testing solutions are presented in Table 2.

The difference between normal concrete (NC) and high performance concrete (HPC) is that HPC utilises cementitious material (e.g. slag, fly ash, silica fume) together with or instead of Portland cement, resulting in the lower alkalinity of its pore solution after hydration. The use of cementitious materials in the concrete is beneficial for reducing the carbon footprint during concrete production process and alleviating concrete expansion caused by alkali silica reaction (ASR). Li et al. [16] used slag instead of Portland cement to produce SWSSC.

To prevent the interaction of carbon dioxide in air with the calcium hydroxide in the simulated concrete solutions, FRPs coupons/samples were placed in the sealed containers where they are fully immersed in the test

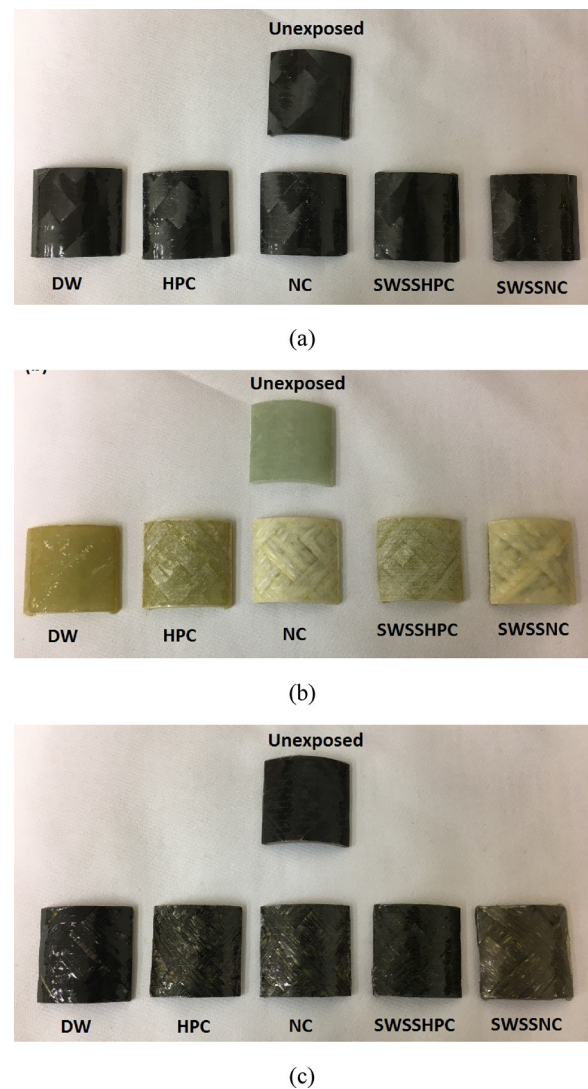


Fig. 1. Surface changes of (a) CFRP (b) GFRP (c) BFRP after 6-month exposure to 5 testing solutions at 60 °C.

Table 2
Chemical composition and pH of test solutions.

Solution No.	Simulated environment	Quantities (g/l)				pH
		NaOH	KOH	Ca(OH) ₂	NaCl	
1 ^a	SWSSNC	2.4	19.6	2.0	35	13.4
2 ^b	NC	2.4	19.6	2.0	–	13.4
3 ^a	SWSSHPC	0.6	1.4	0.037	35	12.7
4 ^b	HPC	0.6	1.4	0.037	–	12.7
5	DW	–	–	–	–	7.5

^a Simulated SWSSNC and SWSSHPC pore solutions were prepared according to [14].

^b NC and HPC simulated pore solutions were prepared according to [6].

Table 1
Chemical compositions of E-glass and basalt fibers.

Types	Chemical composition (wt%)												
	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	CaO	MgO	Na ₂ O	K ₂ O	TiO ₂	MnO	P ₂ O ₅	ZnO	CuO	BaO
E-glass	46.7	10	0.31	17.1	2.09	0.30	0.27	0.23	0.01	0.04	0.01	< 0.01	0.02
Basalt	46.7	14.6	8.51	6.69	2.83	2.07	1.31	0.97	0.14	0.16	0.04	0.01	0.03

solutions (i.e.; simulated concrete solutions, simulated sea water sea sand concrete solutions and DI water). Then the sealed containers are immersed in water baths that are maintained at the required temperature.

2.3. Moisture uptake test

Dry and pre-weighed specimen coupons were immersed in different testing solutions (Table 2) in temperature-controlled water baths set at 25 °C, 40 °C and 60 °C for different durations up to 6 months. Triplicate runs were performed under each condition. After given durations, the coupons were surface dried with paper towel and then weighed using a digital balance to 0.001 g. The weight gain was calculated using the equation below:

$$W_{\text{gain}}(\%) = \frac{W_i - W_0}{W_0} \times 100 \quad (1)$$

where W_0 = pre exposure specimen weight after 24 h oven drying (g)

W_i = post exposure specimen weight (g)

2.4. Scanning electron microscopy (SEM)

Scanning electron microscopy (SEM) observation and image analysis were performed on selected FRP specimens after exposure to different testing solutions using JOEL7001, to characterise morphological changes.

SEM was performed for characterisation of degradation in the areas of fiber, matrix and interface in specimens prepared by two methods: grinding-and-polishing method of the cross section and fracturing after exposure to liquid nitrogen. Grinding and polishing provided a flattened cross section of specimen where degradation at fiber, matrix and their interface region could be observed. One could argue that such grinding could cause mechanical damage to the interface. To address this concern, specimens were exposed to liquid nitrogen that embrittled the FRPs, thereby facilitating brittle fracture along fiber/matrix interface under applied load, and allowing SEM analysis of the interface. Thus, liquid nitrogen fracturing enabled observation of a mechanical damage-free interface. FRP specimens were coated with a thin layer of iridium for the required conductivity to enable SEM imaging.

2.5. Attenuated total reflectance–Fourier transform infrared spectroscopy (ATR-FTIR)

BRUKER-FTIR spectrometer equipped with an attenuated total reflectance was used to study the chemical nature of unexposed FRPs. Three specimens were tested for each FRP for examination of reproducibility.

After 6-month exposure at 60 °C, the residue of each test solution was collected for FTIR analysis, and the obtained spectrum was compared with those of unexposed specimens to determine whether FRP matrix had leached during exposure. In order to obtain residues, a 6-month exposure solution was transferred to a clean and dry container placed in a sterilized hood under a continuous air stream until entire solution evaporated, leaving some residue behind. The residue was mixed with an organic solvent, subjected to ultra-sonication at room temperature for 30 min and held still for 48 h. A few drops of this solution were held on a diamond crystal plate under slow and warm air stream until entire solvent was evaporated. The residue thus obtained was subjected to the FTIR test. The process was repeated three times for each type of exposure solution, and the same FTIR trend in the spectrum was observed for each type. FTIR measurements were carried out in the range of wave numbers from 4000 to 500 cm^{-1} , and 64 scans were acquired with spectral resolution of 4 cm^{-1} .

3. Results and discussion

3.1. Visual observation

As seen from Fig. 1, after 6-month immersion at 60 °C, the colour and luster of exposed CFRP specimens remained largely unchanged,

irrespective of the solutions they were exposed to. On the other hand, the colour of the GFRP and BFRP specimens immersed in simulated concrete solutions (i.e., NC, HPC, SWSSNC, SWSSHPC) turned brighter, and their surface texture became rough, whereas the specimens exposed to distilled water showed only slight colour change with little change in luster. Those surface changes were more prominent in the case of GFRP and BFRP exposed to higher alkalinity solutions (i.e. NC, SWSSNC).

3.2. Moisture uptake

Fig. 2(a)–(c) present the moisture uptake of CFRP, GFRP and BFRP during exposure to different testing solutions (Table 2) at 60 °C for up to 6 months. Moisture uptake results at 40 and 25 °C (room temperature) are presented in Figs. A1 and A2 respectively in Appendix A. For each test solution, the average weight gains of three replications were plotted with normalised immersion time (i.e., immersion days^{1/2}).

As seen in Fig. 2, moisture uptake (weight gain) of each of FRP type increased with immersion time and exposure temperature. However, for immersions in the DW, HPC and SWSSHPC, their weight gains reached a saturation stage after 1 month, while specimens immersed in the higher alkaline environments (i.e. NC, SWSSNC) continued to gain weight at high rate for the entire test duration. In fact, BFRP showed an acceleration in weight gain after ~90 days. This phenomenon of a greater water uptake indicates a more intense degradation in the environment with high alkali content (i.e., NC/SWSSNC). This observation is consistent with the greater loss in tensile strength of FRP bars in high alkaline solutions (i.e. NC, SWSSNC) reported in literature [6,14].

A similar moisture uptake pattern with respect to environment (i.e. NC > SWSSNC > DW > HPC > SWSSHPC) was noticed for all FRPs at all exposure temperatures (Figs. 2, A1 and A2 in Appendix A), with an exception of BFRP at 60 °C (Fig. 2(c)). There is a distinct trend of minimum water uptake by each FRP type in the environment simulating high performance concrete (HPC and SWSSHPC), clearly indicating that role of considerably lower alkali content of such concrete in the resistance of FRPs to degradation/water uptake. There is also an interesting trend of somewhat lesser water uptake when seawater was present with HPC (i.e. HPC > SWSSHPC). Between the weight gain in the environment simulating normal concrete (i.e. NC and SWSSNC), the one with seawater and sea sand (i.e. SWSSNC) was generally lower, again suggesting the beneficial role of the presence of seawater and sea sand in resisting water uptake/degradation. However, BFRP suffered considerable water uptake in SWSSNC than in NC, particularly after 90 days of exposure, indicating deleterious role of NaCl in this case. In order to characterise the degradation of FRPs and their components, SEM was performed on pre-exposed FRP specimens and FTIR was performed on residues of the exposure solutions.

3.3. Fourier transform infrared spectroscopy (FTIR)

Fig. 3 shows the FTIR absorption spectra of unexposed FRPs (i.e. reference) and the residues of the solutions after exposure of FRPs for 6 months at 60 °C. Any leaching of functional groups from specimen resin during the exposure will lead to the appearance of corresponding FTIR bands on the residue spectrum. Table 3 summarizes the detected characteristic absorption bands for the unexposed FRP specimens and their assigned functional groups.

Three groups of FTIR bands can be observed on the reference spectrum for each of the three FRPs. The group located at wavenumber ~ 3000 cm^{-1} represents O–H stretching band, whereas the adjoining bands at 2930–2900 cm^{-1} originate due to stretching vibration of C–H group of epoxy [17,18]. The third group of bands located < 2500 cm^{-1} includes C=C stretching in alkenes and aromatics at ~1610 and ~1509 cm^{-1} , O–H stretching of alcohol at 1415 cm^{-1} , C–N stretching at ~1300 cm^{-1} , asymmetric and symmetric stretching vibration of C–O–Φ at ~1245 and ~1041 cm^{-1} , C–OH at ~1220 cm^{-1} , out of plane bending of C–H group in benzene ring at ~827 cm^{-1} and CH₂ rocking at ~800 cm^{-1} [17–21]. The appearance of C–N band in each spectrum corresponds amine hardener present in the resin.

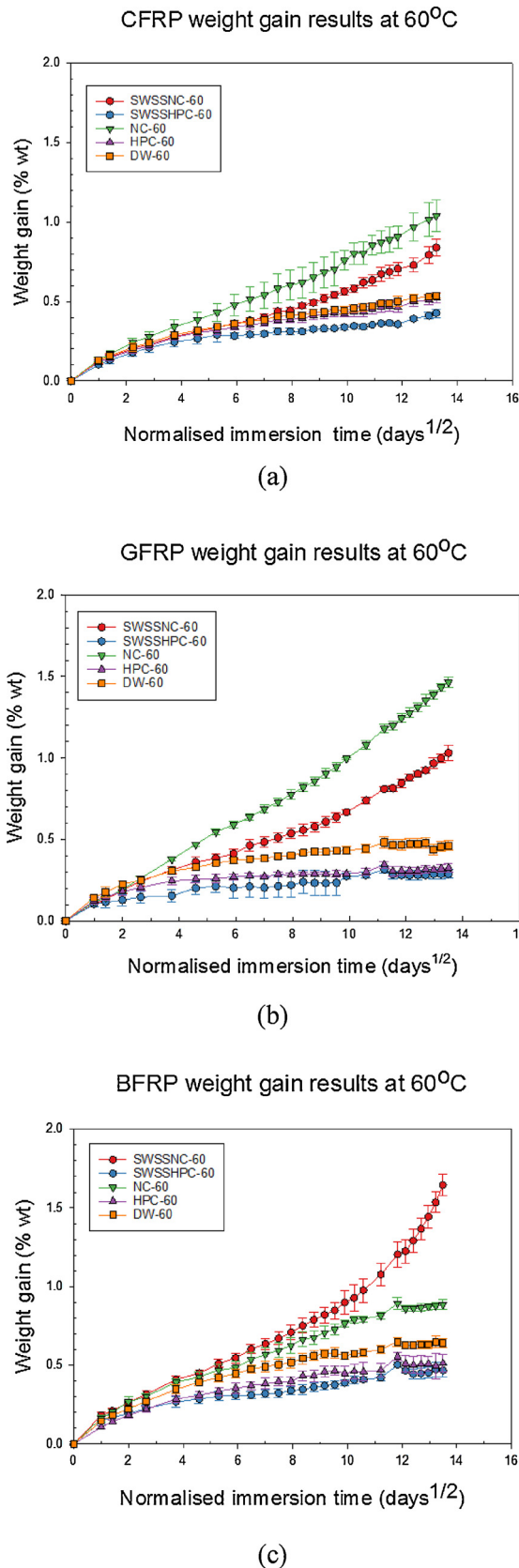


Fig. 2. Moisture uptake of (a) CFRP (b) GFRP (c) BFRP after exposure to different testing solutions at 60 °C for up to 6 months.

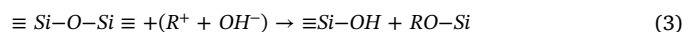
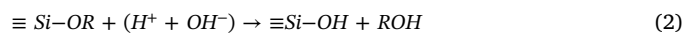
In the case of the spectra for residue, two new bands appeared in addition to the bands for the FRPs described earlier. The new band at $\sim 840 \text{ cm}^{-1}$ corresponds to the changes in the vibration characteristics of the ether group as a result of the water molecules that get embedded in the polymer which also involves hydrogen bond with the C–O–C groups [17]. The second band, i.e. at $\sim 1125 \text{ cm}^{-1}$, can be attributed to the changes in the vibration characteristic of C–OH band due to hydrogen bond (C–O–H...OH₂) during hydration [17]. Most bands similar of the reference spectra for FRPs also appeared on the residue spectrum, indicating the leaching of composite resin during exposure. A greater intensity of the O–H stretching peak of alcohol at $\sim 1415 \text{ cm}^{-1}$ was observed on the residue spectrum of NC and SWSSNC but only for BFRPs exposed to SWSSHPC, which may indicate a greater embedding of water (or hydroxyl group) in resin during the immersion in these cases [17].

3.4. Scanning electron microscopy (SEM)

Fig. 4 shows the SEM images of the cross section of area around the edge of the BFRP specimens that were exposed to simulated SWSSHPC solution at 60 °C for 6 months. The fiber/matrix interface is seen to have suffered debonding and some fracture. Such features were observed for each of FRP types exposed to different test environments (Fig. 4b). The debonded and fractured interfaces may join, giving an appearance of continuous crack. In the case of the BFRP and GFRP specimens exposed to NC/SWSSNC (Fig. 5(a) & (b)), the development of such continuous cracks possibly provided more space and pathways, and facilitated additional moisture uptake by FRPs exposed to the two solutions. The considerably higher alkali contents of NC and SWSSNC can degrade not only the resin of FRPs, it can also attack fibers, particularly those in BFRP and GFRP, accelerating the damage at the fiber/matrix interface, as seen in Fig. 5(a) and (b). In contrast, cross section of the pre-exposed CFRP showed very limited damage (Fig. 5(c)).

However, one could argue whether the damages seen at the fiber/matrix/interfaces in BFRP and GFRP were actually caused during the grinding. To dispel this ambiguity, the specimens of FRPs pre-exposed to SWSSNC for 6 months at 60 °C were immersed in liquid nitrogen, and then quickly fracture upon sudden loading. The exposure to liquid nitrogen embrittled the resin matrix, and the subsequent loading caused fracture along the fiber/matrix interface and produced mechanical damage-free surface for SEM imaging. Indeed, extensive fiber degradation was observed, particularly in the edge area of GFRP and BFRP exposed to each of the simulated concrete environment (i.e. NC, SWSSNC, HPC, SWSSHPC), and observed degradation features included pits, plate-shaped corrosion products, ring-shaped cracks and corrosion shells with a porous structure and amorphous surface (Figs. 6–8). However, in higher alkaline environment (i.e. NC, SWSSNC), fiber degradation in BFRP and GFRP was more extensive that extended deeper into the composite specimens, indicating the considerably high alkali content had reacted aggressively at 60 °C with fibers. Such fiber degradation facilitated the interface degradation that allowed greater water penetration, thereby accounting for the greater moisture uptake of FRP composites in such simulated environments.

Corrosion shell and pitting are reported to be degradation features of both sized or unsized glass/basalt fibers in alkaline environment [22–25]. Normally, these features can be observed on glass/basalt fibers just after a few days or a few hours of immersion in alkaline solutions even at normal temperature. Herein, the degradation of fibers of FRPs was delayed since they were protected by the surrounding matrix. Previous studies suggested that the formation of corrosion shell on glass/basalt fibers in alkaline environments is the result of the reaction of alkali-ions with silicate in fibers (Eq. (2)), and subsequent destruction and gradual dissolution of the silicate network as presented in Eq. (3) [23–27].



The thickness of corrosion shell increases with time, and the corrosion front gradually moves towards the fiber core, leaving behind the insoluble

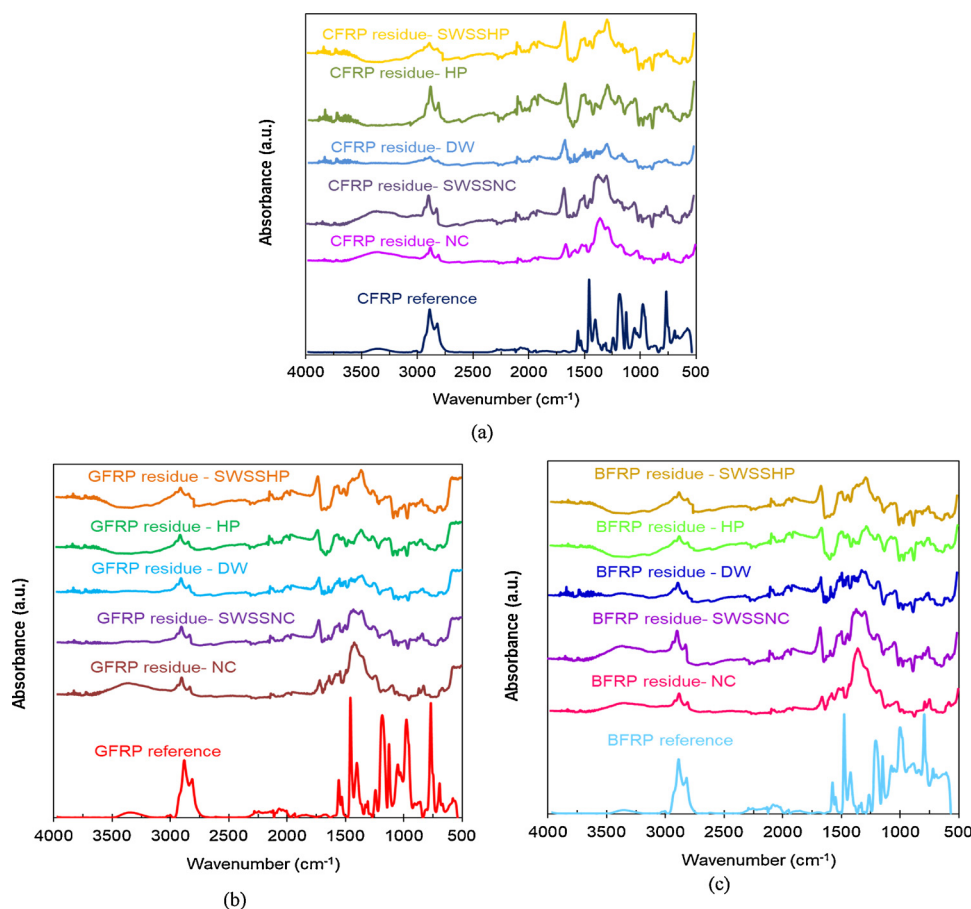


Fig. 3. FTIR absorption spectra for the reference specimens and the solution residues after exposure of (a) CFRP (b) GFRP (c) BFRP at 60 °C for 6 months.

Table 3

Characteristic absorption bands observed on unexposed FRP specimens.

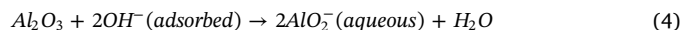
Absorption bands (cm ⁻¹)	Assignment
3400	O–H stretching band
~2930 and ~2900	Stretching vibration of C–H group
~1610	C=C stretching band (alkene)
~1509	C=C (aromatic nucleus)
~1415	O–H stretching (alcohol)
~1300	C–N stretching band
~1245	Asymmetric stretching vibration C–O–Φ
~1220	C–OH (of guaiacyl ring)
~1040	Symmetric stretching vibration C–O–Φ
~827	Out of plane bending of C–H (benzene)
~800	CH ₂ rocking

compounds of calcium, iron, titanium, magnesium and zirconium [23]. As the corrosion shell is expansive, its growth induces local stress that eventually leads to interface debondings and exfoliation from fibers. The exfoliation is facilitated by the mismatch of mechanical properties between the corrosion shell and fiber. The attack could also be localised, such as the formation of the pits (Figs. 6–8). A pit reduces the local diameter of the fiber, compromising its load-bearing ability.

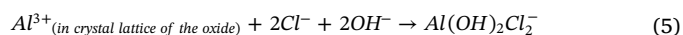
Though fibers and fiber/matrix interfaces of both BFRP and GFRP suffered extensive corrosion in conventional concrete as well as SWSSC, the extent of attack was considerably more in the case of BFRP (Fig. 7(a)). The greater extent of fiber and fiber/matrix interface degradations of BFRP is consistent with the greater moisture uptake of BFRP as seen in Fig. 2. Wang et al. [14] have also reported greater loss of tensile strength of BFRP than GFRP after 6-month exposure to SWSSNC solutions. This difference could result from a chemical reaction between aluminium, iron and magnesium contents of fibers (Table 1) with alkali and chloride ions in the simulated

concrete environments. Table 4 presents the chemical compositions of the simulated solutions of normal concrete (NC) and seawater sea sand normal concrete (SWSSNC), after six months of immersion of BFRP in these simulated environments at 60 °C. As a result of immersion of BFRP for 6 months, the test solution developed considerable contents of Al, Fe and Mg whereas the contents of other elements (Mn, Ti, P, Zn, Cu, Cr, Mo and Ba) were insignificant in simulated solutions under investigation. Na, K and Ca are excluded from the comparison as they are present in the original simulated concrete solutions. The presence of Si has already been explained earlier.

In alkaline environment (pH > 9), aluminium oxide dissolves according to the reaction [28,29]:



Chloride ions from SWSSC react with aluminium ions, and form soluble oxychloride complexes ($\text{Al}(\text{OH})_2\text{Cl}_2^-$) [30]:



Thus the soluble $\text{Al}(\text{OH})_2\text{Cl}_2^-$ complexe is leached into the simulated concrete solutions. Similarly, the Mg oxide/hydroxide also dissolves in the solution containing chloride ions [28].

With respect to the iron in basalt fiber the proposed reaction mechanism can be summarised by the following steps:

- i) In an aqueous environment (natural water environment), ferric hydroxide, $\text{Fe}(\text{OH})_3$ forms upon hydrolysis of ferric iron [31]:



- ii) The rate of hydrolysis of the ferric hydroxide increases in alkaline solution as its solubility is pH dependent [32].
- iii) When chloride ions are also present, ferric iron can react with them

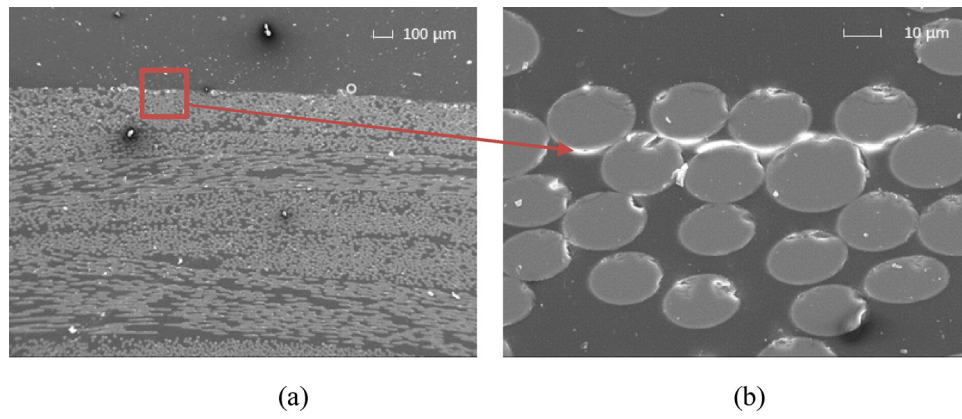


Fig. 4. SEM images for BFRP immersed in SWSSHPC at 60 °C for 6 months (a) magnification X50 (b) magnification X1000.

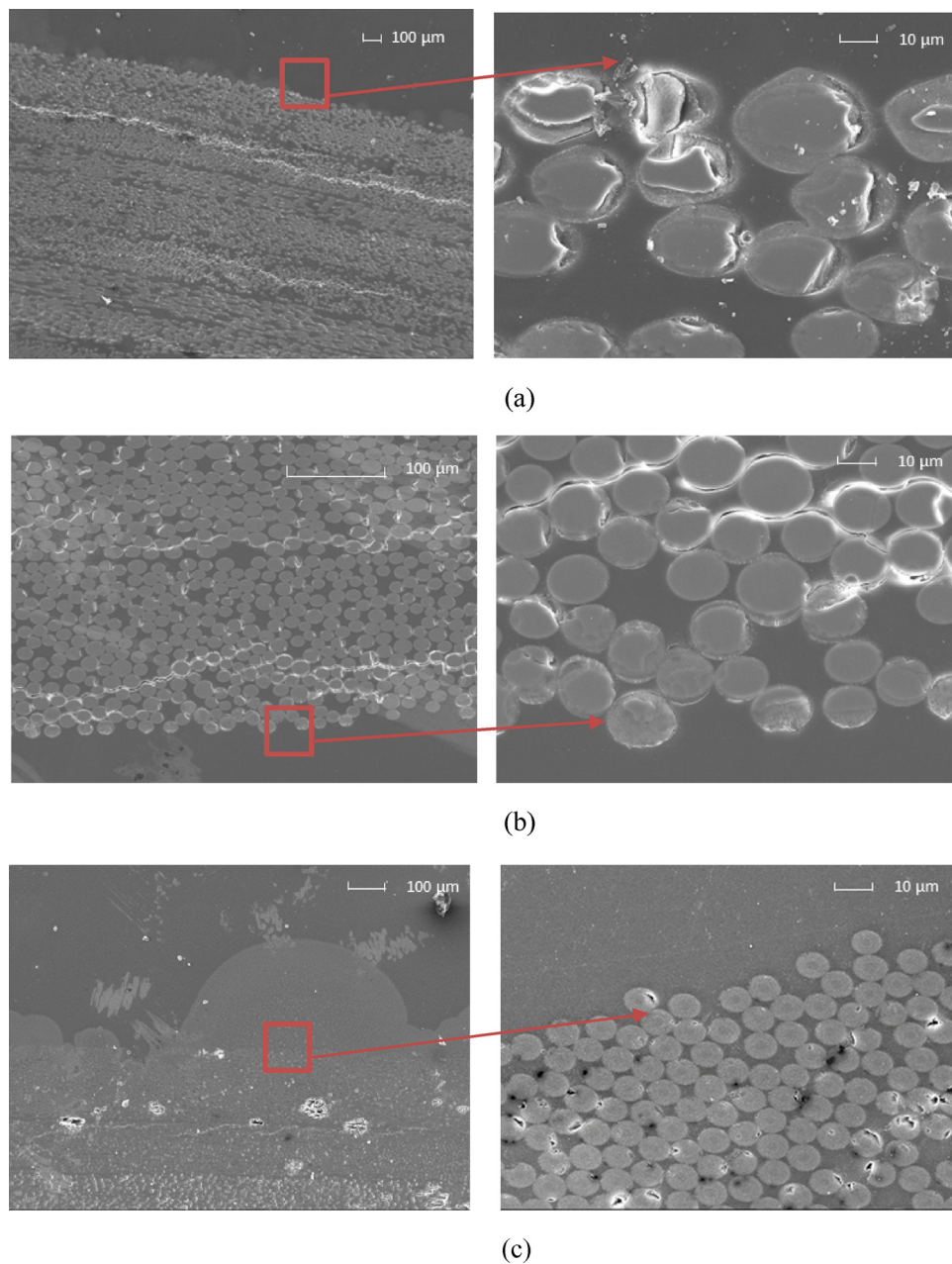


Fig. 5. SEM images for FRP immersed in SWSSNC at 60 °C for 6 months (a) GFRP (b) BFRP (c) CFRP.

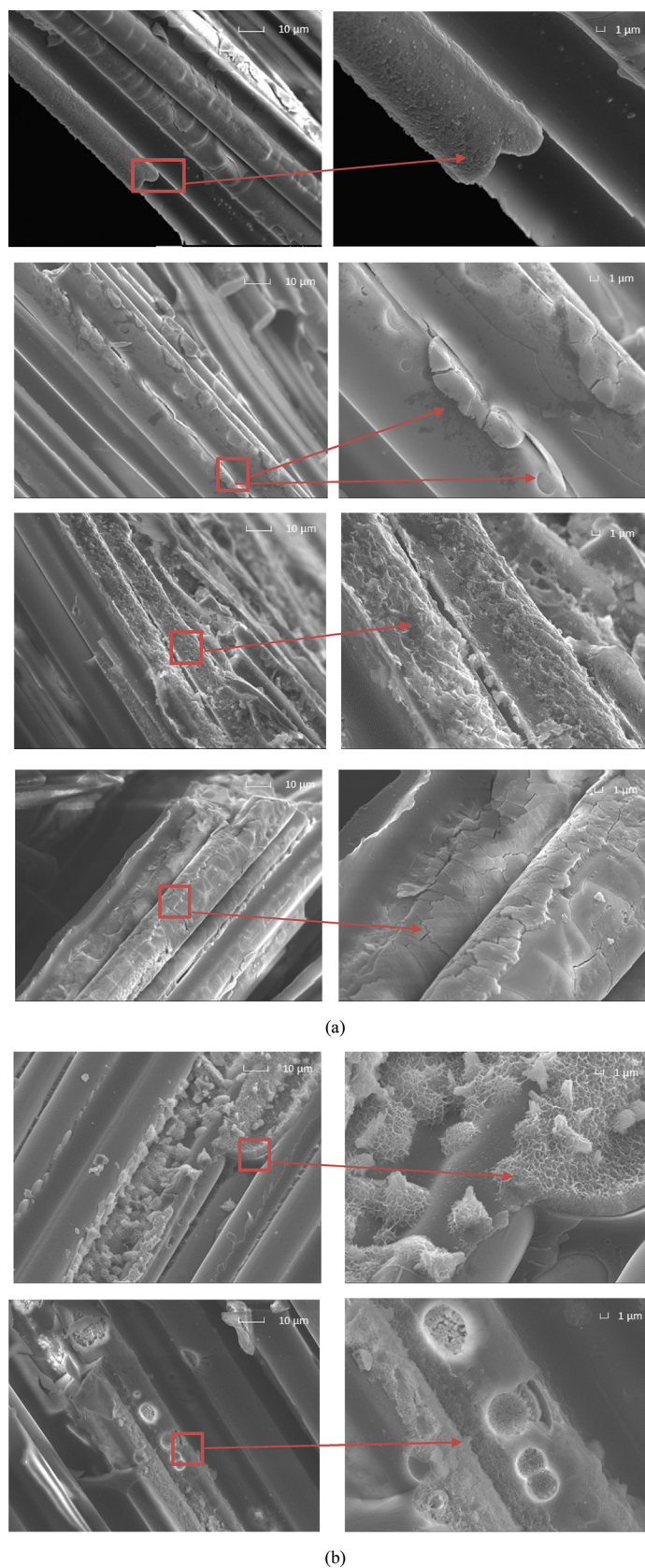


Fig. 6. SEM images of (a) BFRP (b) GFRP exposed to SWSSNC at 60 °C for 6 months.

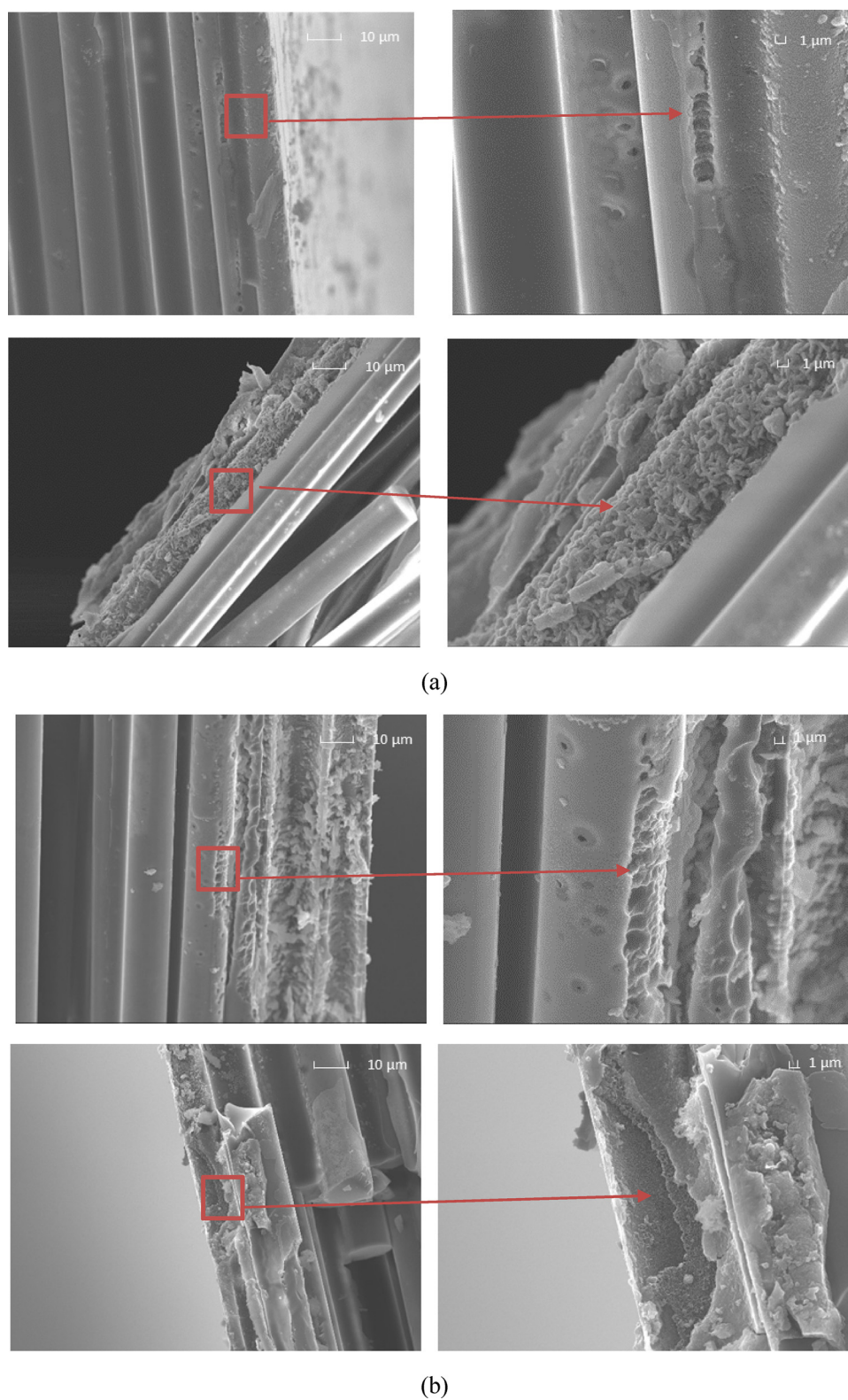


Fig. 7. SEM images of (a) BFRP (b) GFRP exposed to SWSSHPC at 60 °C for 6 months.

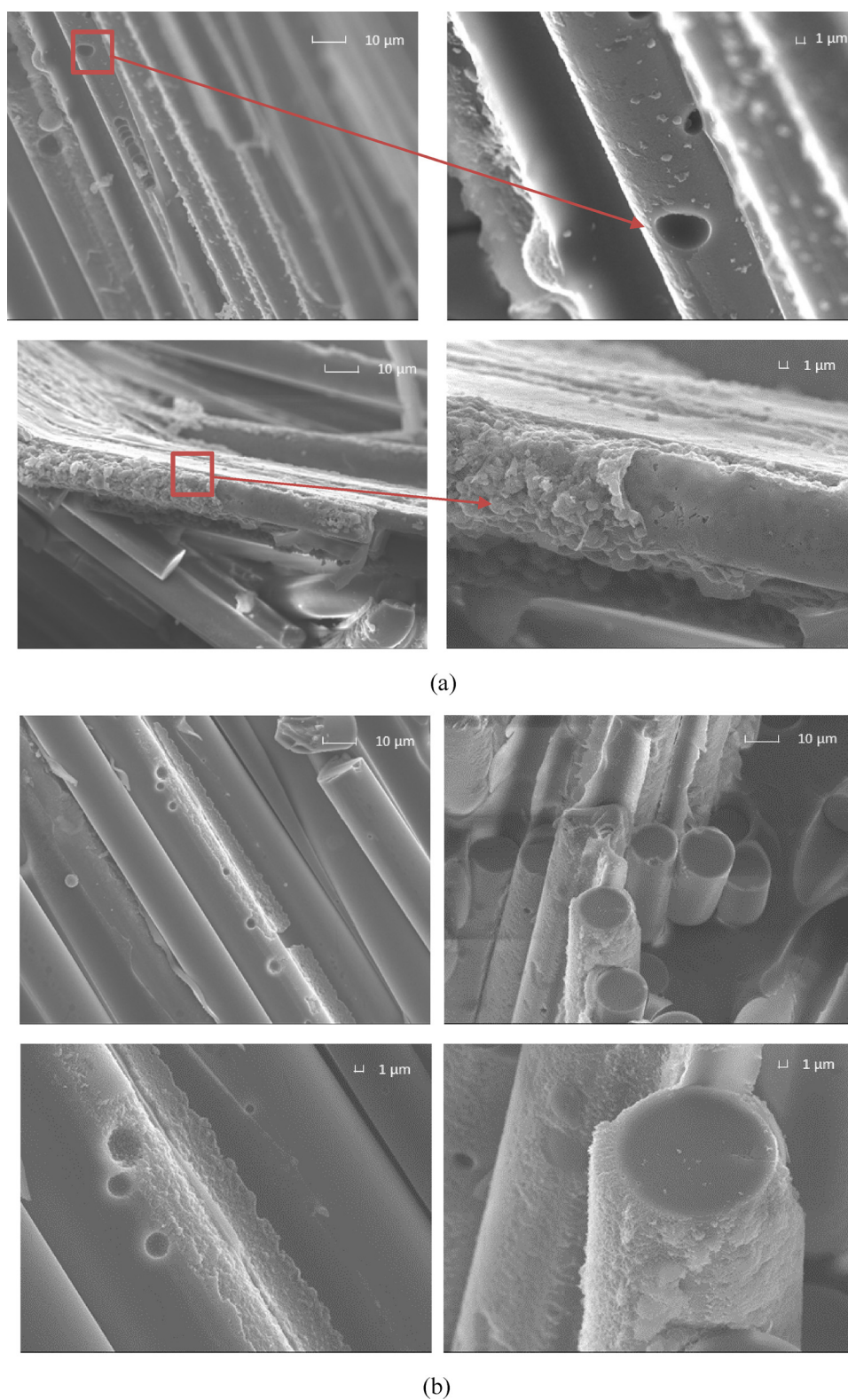
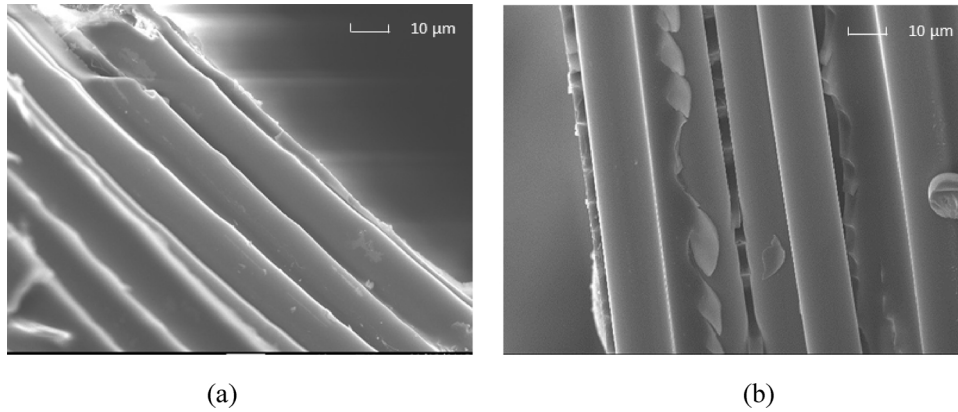


Fig. 8. SEM images of (a) BFRP (b) GFRP exposed to NC at 60 °C for 6 months.

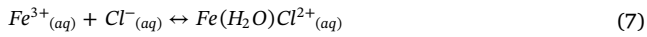
Table 4

Chemical composition of the simulated solutions of normal concrete (NC) and seawater sea sand normal concrete (SWSSNC) after six months of immersion of BFRP at 60 °C.

Types	Chemical composition (mg/l)												
	Si	Al	Fe	Ca	Mg	Na	K	Ti	Mn	P	Zn	Cu	Ba
NC	71	11	2	1325	7	3450	21045	< 1	< 1	< 1	< 1	< 1	< 1
SWSSNC	94	16	3	1430	8	17100	17880	< 1	< 1	< 1	< 1	< 1	< 1

**Fig. 9.** SEM images of (a) BFRP (b) GFRP exposed to DW at 60 °C for 6 months.

as shown in the following equations, where $Fe(H_2O)Cl^{2+}$ is the iron-chloride complex [33].



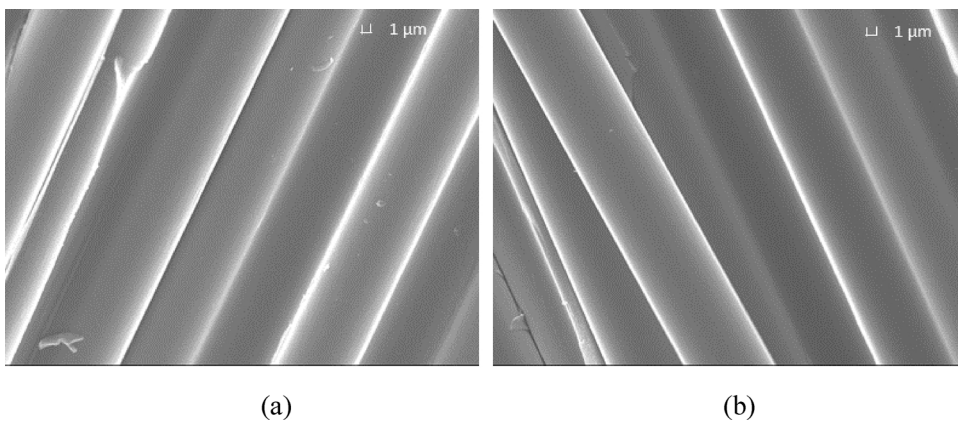
It was reported in [32] that the osmotic coefficient in the presence of NaCl solution is a function of $FeCl_2$ molal concentration, which explains significant water uptake of BFRP in SWSSNC environment (Fig. 2c), because basalt fibers exclusively have iron oxide content (Table 1).

In contrast, no significant degradation was observed on BFRP and GFRP exposed to distilled water (Fig. 9). In regards to CFRP, its fibers and fiber/matrix interface were found to be smooth and clean even after exposure to NC/SWSSNC at 60 °C for the same duration, as shown in Fig. 10, confirming that irrespective of the exposure solution, carbon fibers largely remained unattacked after exposure.

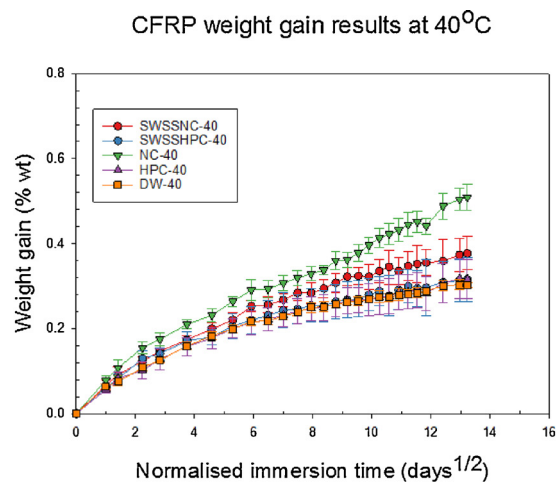
4. Conclusion

This study investigated the degradation behaviour of three types of filament wound FRPs (i.e. CFRP, GFRP, BFRP) exposed to simulated SWSSC and other concrete solutions. Key findings can be summarised as following:

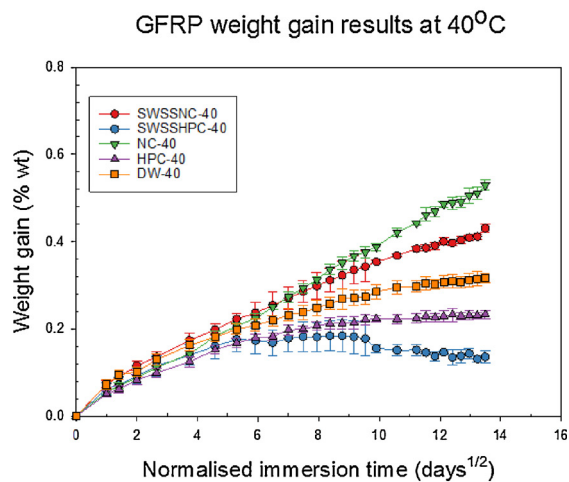
- 1 Moisture uptake of FRPs (except BFRP at 60 °C) during exposure to various environments followed a trend: NC > SWSSNC > DW > HPC > SWSSHPC
- 2 Degradation of resin and fibers, particularly basalt and glass fibers, in the simulated high performance concrete solutions that have considerably lower alkali contents was considerably lower.
- 3 The presence of NaCl in simulated concrete environment has a beneficial role for FRPs in resistance to moisture uptake with an exception of BFRP at 60 °C. The exception of BFRP at 60 °C is attributed to a chemical reaction between the aluminium, iron and magnesium contents of BFRP with chlorides.
- 4 CFRP has the best durability performance in simulated SWSSC environments, whereas the fiber and fiber/matrix interface degradation on BFRP and GFRP in such environments, especially in higher alkalinity solution (i.e. SWSSNC), suggests the need to accurately predict its long-term performance.

**Fig. 10.** SEM images for 6-month exposed CFRP fractured in liquid N₂ (a) SWSSNC (b) NC.

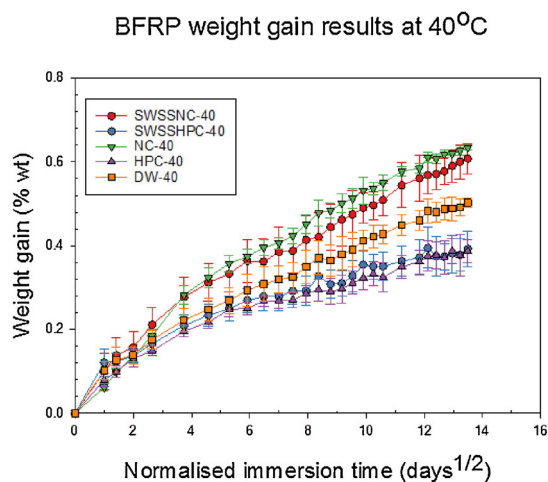
Appendix A



(a)

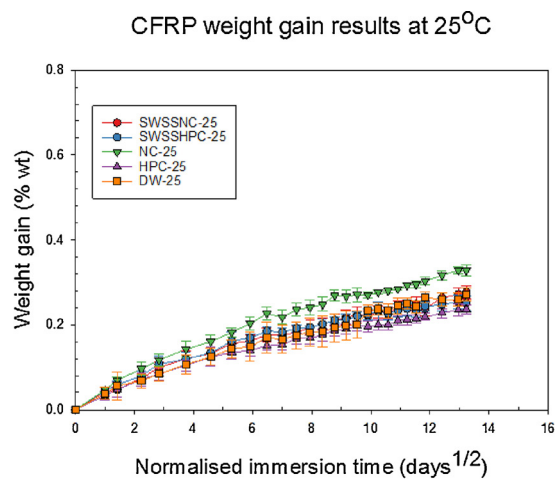


(b)

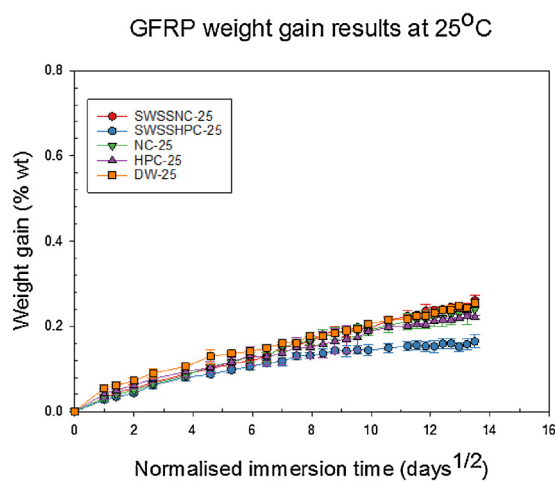


(c)

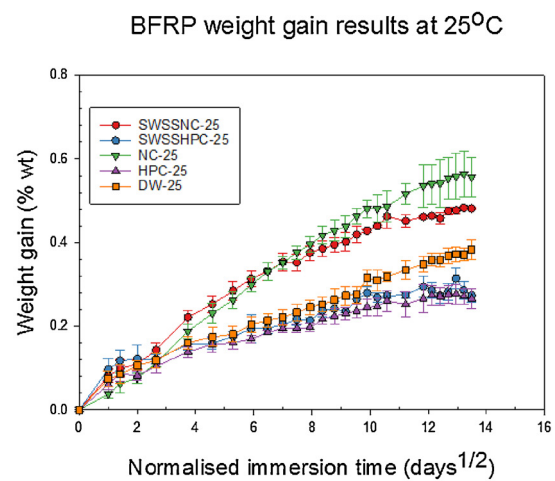
Fig. A1. Moisture uptake of (a) CFRP (b) GFRP (c) BFRP after exposure to different testing solutions at 40 °C for up to 6 months.



(a)



(b)



(c)

Fig. A2. Moisture uptake of (a) CFRP (b) GFRP (c) BFRP after exposure to different testing solutions at 25 °C (room temperature) for up to 6 months.

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